Power

File: power.kicad_sch

MCU

File: mcu.kicad_sch

USB_Jetson

File: usb_jetson.kicad_sch

SmartPort_A

File: smartport.kicad_sch

Sensors

File: sensors.kicad_sch

IMU

File: imu.kicad_sch

SmartPort_B

File: smartport.kicad_sch

VIN_OR

VBAT_OR

TX

RXD

DE

RS485A_TX

RS485A_RX

RS485A_DE

RS485B_TX

RS485B_RX

RS485B_DE

Port A and port B each have their own MAX33071E (smartport sheet).
Firmware can split them, A to read from the brain and B to write to it, or run both half duplex.

H1 #8-32

H2 #8-32

H3 #8-32

H4 #8-32

H1 to H4: #8-32 screws into the two brain mount brackets (brainMount-frontSlant left and right).
Hole centres 93.92 mm (3.6976 in) left to right, 50.80 mm (2.000 in) top to bottom.
4.5 mm non-plated holes, slightly larger than the 0.175 in (4.45 mm) bracket holes. Not tied to GND.

UWAT VEXU

Sheet: /

File: centurion.kicad_sch

Title: Centurion Sensor Board

Size: A4	Date:	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 1/8

VIN_ORD
VIN_OR

LED1
Green
LED1_A
GND

VIN_OR
R17
10k
LED1_A

VIN_OR
F1
1.5A hold
VIN_MOD

D4
SMBJ16A
VIN_MOD
GND

VIN_MOD
C1
47u 50V
GND

VIN_MOD
C2
10u 50V
GND

VIN_MOD
C3
10u 50V
GND

VIN_MOD
C8
100n
GND

+5V
R18
2.2k
LED2_A

LED2
Green
LED2_A
GND

+5V
R30
10k
FB

FB
R29
2.49k
GND

U1
TPSM53602
VIN_MOD
1 VIN
2 EN
7 VOUT
9 FB
6 PGOOD
13 VSV
12 AGND
15 PGND
+5V
FB
PGOOD
GND

+5V
C4
47u 10V
GND

+5V
C5
47u 10V
GND

J16
USB-C
VBUS
A4 VBUS_USB
CC1
A5 CC1
CC2
B5 CC2
D-
A7 USB_DN
D-
B7 USB_DN
D+
A6 USB_DP
D+
B6 USB_DP
SBU1
A8
SBU2
B8
SHIELD
SH
GND
A1
GND
GND

CC1
R31
5.1k
GND

CC2
R37
5.1k
GND

VBUS_USB
U13
USBLC6-2SC6
USB_DP
USB_DN
USB_DP
USB_DN
GND

VBUS_USB
D10
SS14
USB_PWR

USB_DP
R38
22
USB_D+

USB_DN
R39
22
USB_D-

USB_PWR
Q1
AO3401A
+5V
5V_LDO

D9
SS14
+5V
5V_LDO

5V_LDO
C16
10u 50V
GND

5V_LDO
U7
MCP1826S-3302E
VI
GND
VO
+3.3V
+3.3V

+3.3V
C17
47u 10V
GND

VIN_MOD
PWR_FLAG
R1
100k
VBAT_SENSE
R40
22k
C6
100nF
PWR_FLAG
GND

+3.3V
R45
10k
PGOOD

VBUS_USB
R46
10k
VBUS_SENSE

VBUS_SENSE
R47
15k
GND

VBUS_USB
R48
5.1k
GND

Test pads from powerJackal: VIN_MOD, 5V, 3V3, 2x GND, PGOOD.

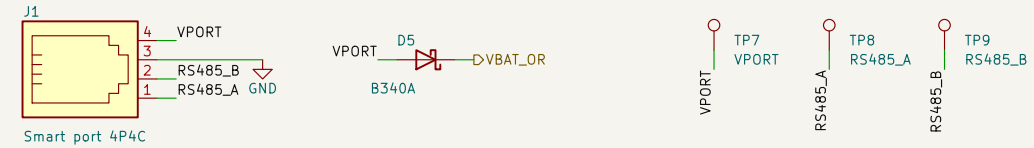
TP1
VIN_MOD
TP2
5V
TP3
3.3V
TP4
GND
TP5
GND
TP6
PGOOD
PWR_FLAG
5V_LDO

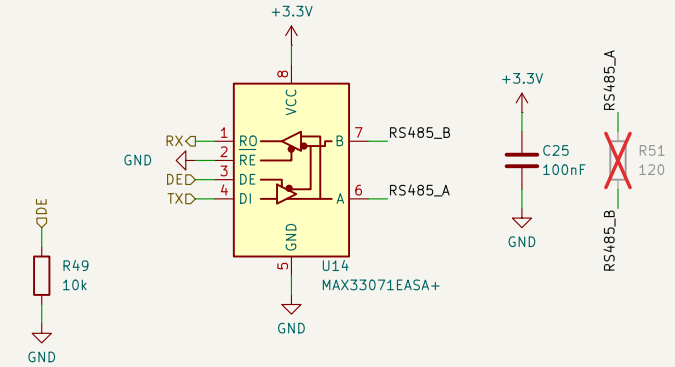
VBUS_USB
C30
100nF
GND

VIN_MOD
C32
10u 50V
GND

R45: PGOOD pullup (open drain), read on PC3.
On USB power alone the buck is off and PGOOD may
still read high: firmware checks VBAT_SENSE instead.
R46/R47: VBUS sense, 5 V gives 3.0 V on PB2.
Connect USB D+ only while VBUS_SENSE is high.
R48: holds VBUS at 0 V with no host plugged in
(same fix as stellaNova R22).

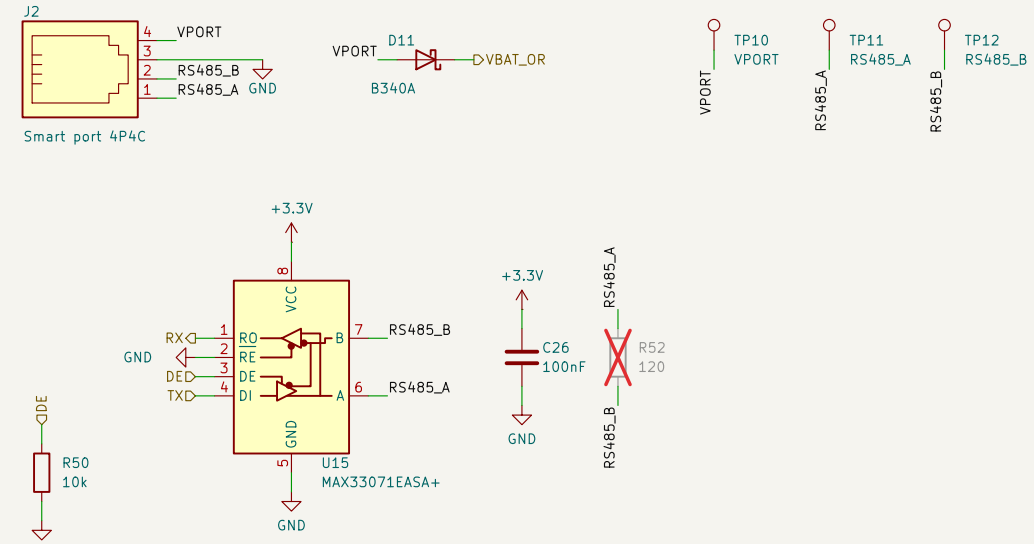
UWAT VEXU
Sheet: /Power/
File: power.kicad_sch
Title: Centurion Sensor Board
Size: A3
Date:
Rev: A
KiCad E.D.A. 10.0.8
Id: 2/8





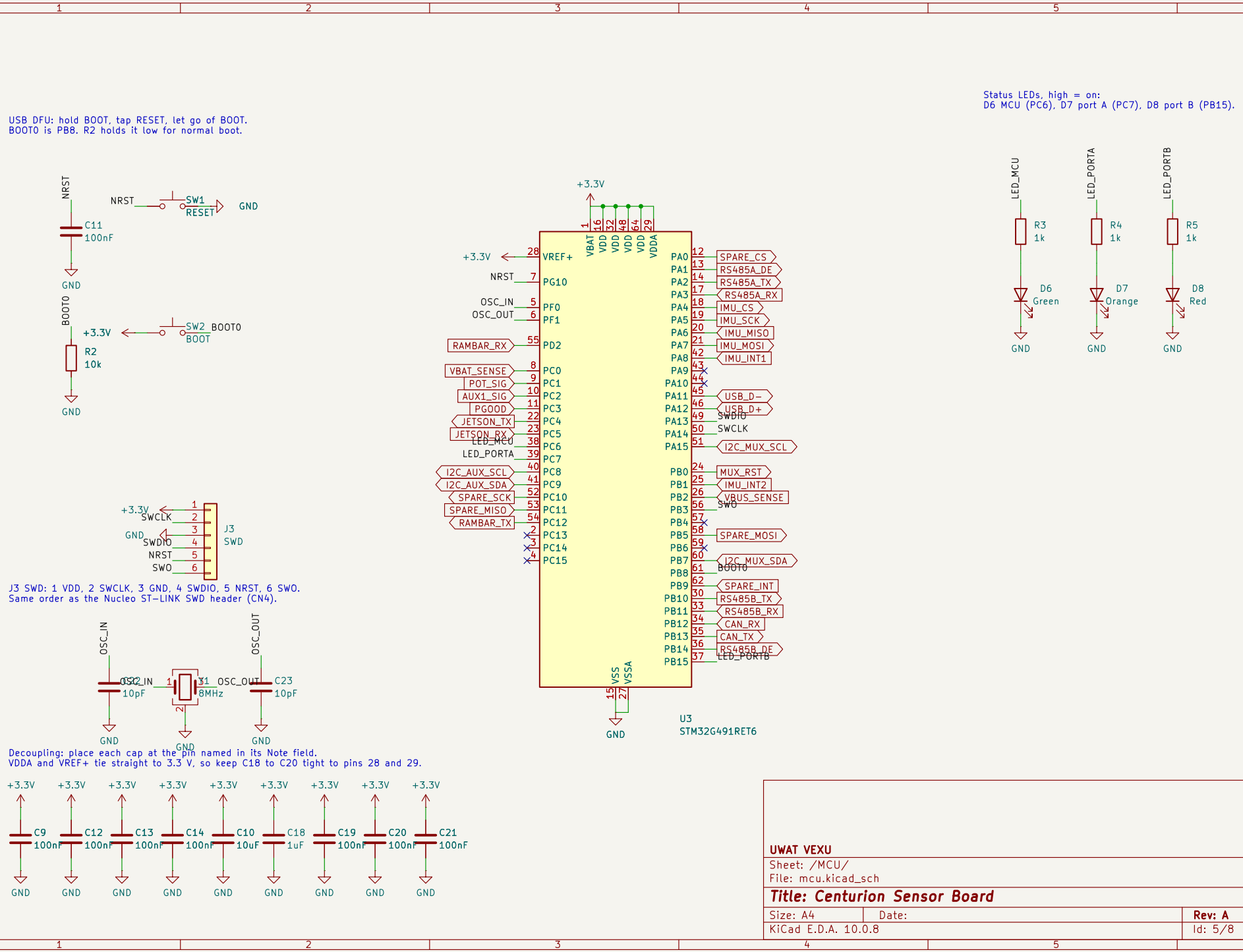
RS-485 transceiver for this port (port A: U14, R49, R51, C25. port B: U15, R50, R52, C26).
DE pulldown keeps the driver off while the MCU resets or is being flashed.
R51/R52 120 ohm termination is not fitted: the smart cable is short. Fit it only if the bus rings.
No TVS on A/B: anything that clamps below 14.6 V would conduct if the battery touched A or B.
The MAX33071E bus pins survive +/-65 V.

UWAT VEXU		
Sheet: /SmartPort_A/		
File: smartport.kicad_sch		
Title: Centurion Sensor Board		
Size: A4	Date:	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 3/8



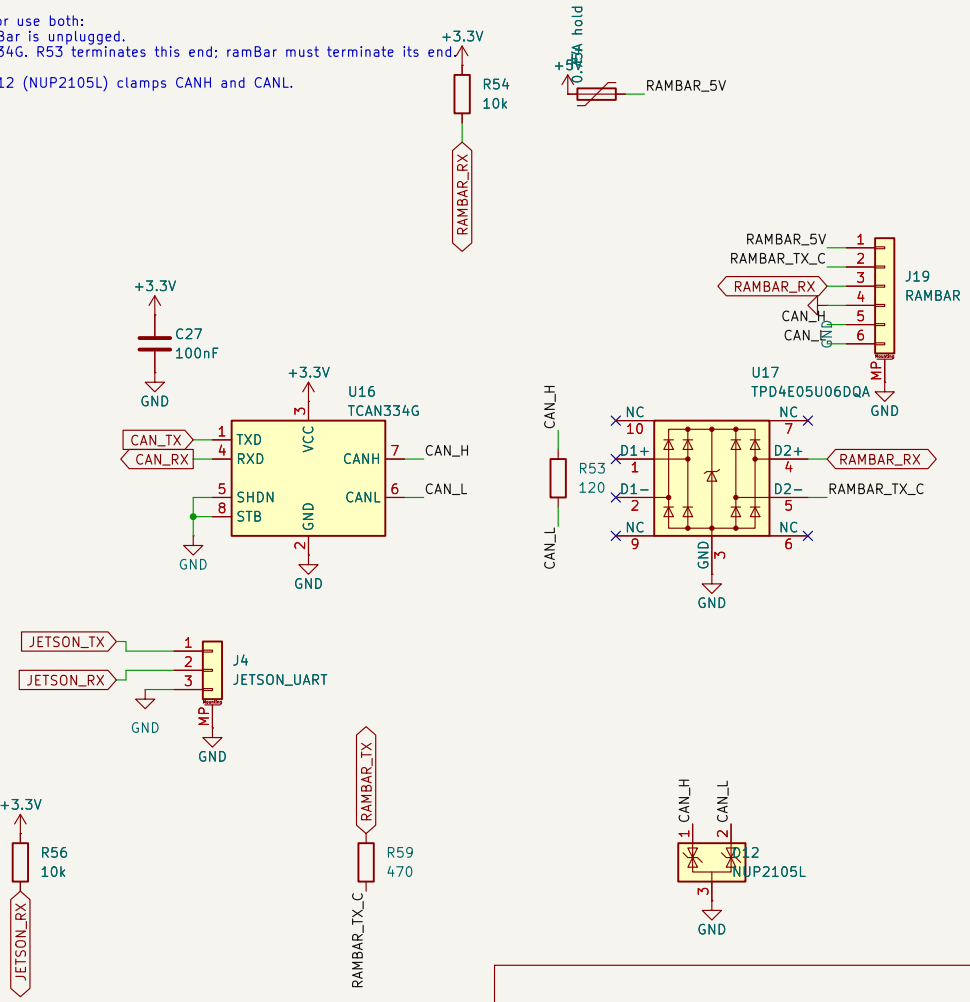
RS-485 transceiver for this port (port A: U14, R49, R51, C25. port B: U15, R50, R52, C26).
DE pulldown keeps the driver off while the MCU resets or is being flashed.
R51/R52 120 ohm termination is not fitted: the smart cable is short. Fit it only if the bus rings.
No TVS on A/B: anything that clamps below 14.6 V would conduct if the battery touched A or B.
The MAX33071E bus pins survive +/-65 V.

UWAT VEXU		
Sheet: /SmartPort_B/		
File: smartport.kicad_sch		
Title: Centurion Sensor Board		
Size: A4	Date:	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 4/8

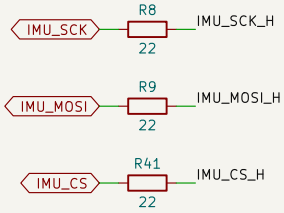

USB DFU: hold BOOT, tap RESET, let go of BOOT.
BOOT0 is PB8. R2 holds it low for normal boot.
NRST
SW1
RESET
GND
C11
100nF
BOOT0
+3.3V
SW2_BOOT0
BOOT
R2
10k
J3 SWD: 1 VDD, 2 SWCLK, 3 GND, 4 SWDIO, 5 NRST, 6 SWO.
Same order as the Nucleo ST-LINK SWD header (CN4).
J3
SWD
SWCLK
SWDIO
SWO
OSC_IN
OSC_OUT
Y1
8MHz
C22
10pF
C23
10pF
Decoupling: place each cap at the pin named in its Note field.
VDDA and VREF+ tie straight to 3.3 V, so keep C18 to C20 tight to pins 28 and 29.
C9
C12
C13
C14
C10
C18
C19
C20
C21
100nF
10uF
1uF
U3
STM32G491RET6
VREF+
PG10
PF0
PF1
PD2
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PC14
PC15
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB11
PB12
PB13
PB14
PB15
VBAT
VDD
VDDA
VSS
VSSA
SPARE_CS
RS485A_DE
RS485A_TX
RS485A_RX
IMU_CS
IMU_SCK
IMU_MISO
IMU_MOSI
IMU_INT1
USB_D-
USB_D+
I2C_MUX_SCL
MUX_RST
IMU_INT2
VBUS_SENSE
SPARE_MOSI
I2C_MUX_SDA
SPARE_INT
RS485B_TX
RS485B_RX
CAN_RX
CAN_TX
RS485B_DE
RAMBAR_RX
VBAT_SENSE
POT_SIG
AUX1_SIG
PGOOD
JETSON_TX
JETSON_RX
LED_MCU
LED_PORTA
LED_PORTB
I2C_AUX_SCL
I2C_AUX_SDA
SPARE_SCK
SPARE_MISO
RAMBAR_TX
Status LEDs, high = on:
D6 MCU (PC6), D7 port A (PC7), D8 port B (PB15).
R3
R4
R5
1k
D6
Green
D7
Orange
D8
Red
UWAT VEXU
Sheet: /MCU/
File: mcu.kicad_sch
Title: Centurion Sensor Board
Size: A4
Date:
Rev: A
KiCad E.D.A. 10.0.8
Id: 5/8

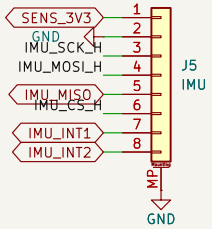
+3.3V
F2
0.35A hold
SENS_3V3
+3.3V
C15
100nF
R12
4.7K
R11
4.7K
R10
10K
I2C_MUX_SCL
I2C_MUX_SDA
MUX_RST
U6
TCA9548A
SCL
SDA
RESET
A2
A1
A0
VCC
GND
SC0
SD0
SC1
SD1
SC2
SD2
SC3
SD3
SC4
SD4
SC5
SD5
SC6
SD6
SC7
SD7
ENC1_SCL
ENC1_SDA
ENC2_SCL
ENC2_SDA
LASER_SCL
LASER_SDA
TOF1_SCL
TOF1_SDA
TOF2_SCL
TOF2_SDA
TOF3_SCL
TOF3_SDA
TOF4_SCL
TOF4_SDA
MUX7_SCL
MUX7_SDA
GND
TOF1_VCC
R19
2.7k
R20
2.7k
TOF1_SCL
TOF1_SDA
J9
TOF1
MP
GND
ENC1_VCC
R13
2.7k
R14
2.7k
ENC1_SCL
ENC1_SDA
J6
ENC1
ENC2_VCC
R15
2.7k
R16
2.7k
ENC2_SCL
ENC2_SDA
J7
ENC2
MUX7_VCC
R27
2.7k
R28
2.7k
MUX7_SCL
MUX7_SDA
J13
SPARE
TOF4_VCC
R25
2.7k
R26
2.7k
TOF4_SCL
TOF4_SDA
J12
TOF4
TOF3_VCC
R23
2.7k
R24
2.7k
TOF3_SCL
TOF3_SDA
J11
TOF3
TOF2_VCC
R21
2.7k
R22
2.7k
TOF2_SCL
TOF2_SDA
J10
TOF2
U8
TPD4E05U06DQA
NC
D1+
D1−
D2+
D2−
U9
TPD4E05U06DQA
LASER_SCL
LASER_SDA
TOF1_SCL
TOF1_SDA
U10
TPD4E05U06DQA
TOF2_SCL
TOF2_SDA
TOF3_SCL
TOF3_SDA
U11
TPD4E05U06DQA
TOF4_SCL
TOF4_SDA
MUX7_SCL
MUX7_SDA
U12
TPD4E05U06DQA
I2C_AUX_SCL
I2C_AUX_SDA
AUX1_SIG_IN
POT_IN
AUX1_VCC
J14
AUX1
JP1
AUX1_VCC
SENS_3V3
SENS_5V
I2C_AUX_SDA
I2C_AUX_SCL
AUX1_SIG_IN
AUX1_SIG
R32
2.2k
R33
2.2k
R34
10k
R36
0k
R35
15k
+5V
F3
SENS_5V
Mux channel 2: lift laser on the turret (J8).
ESD is U9 D1+/D2+.
J17 turret pot: 1 3V3, 2 wiper, 3 GND.
Powered from 3.3 V so the wiper stays inside the ADC range.
1k + 100 nF filter into PC1 (ADC). ESD is U12 D2−.
J18 spare I2C on the second bus (I2C3, PC8/PC9),
shared with AUX1. Pullups R32/R33. ESD U12.
LASER_VCC
LASER_SCL
LASER_SDA
J8
LASER
R42
2.7k
R43
2.7k
POT_VCC
POT_IN
J17
POT
R44
1k
POT_SIG
C24
100nF
I2C_SPARE_VCC
J18
I2C_SPARE
2 ENC1_VCC
2 ENC2_VCC
2 LASER_VCC
2 TOF1_VCC
2 TOF2_VCC
2 TOF3_VCC
2 TOF4_VCC
2 MUX7_VCC
2 I2C_SPARE_VCC
JP2
JP3
JP4
JP5
JP6
JP7
JP8
JP9
JP10
Sensor jack supply select (JP2–JP10). Each jack has its own VCC net; 1x3 2.54 mm header + shunt:
pins 1–2 = 3.3 V, pins 2–3 = 5 V, no shunt = jack unpowered. JP1 (AUX1) works the same way.
Mux jacks: the pullups follow the jack's VCC, and the TCA9548A translates 3.3 V <–> 5 V per channel.
J18 (bus 2, straight to the MCU): only VCC changes, the bus pullups R32/R33 stay on 3.3 V.
F3 fuses the 5 V sensor branch the same way F2 fuses SENS_3V3. AUX1 (JP1) also takes 5 V from SENS_5V.
Mux channels: 0 ENC1 (J6), 1 ENC2 (J7), 2 LASER (J8), 3–6 TOF1–TOF4 (J9–J12), 7 SPARE (J13).
+3.3V
R55
100
POT_VCC
C28
10uF
C29
10uF
R60
1M
C31
10nF
UWAT VEXU
Sheet: /Sensors/
File: sensors.kicad_sch
Title: Centurion Sensor Board
Size: A3
Date:
Rev: A
KiCad E.D.A. 10.0.8
Id: 6/8

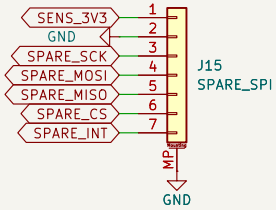
ramBar link (J19, JST-GH 6 pin vertical). Pick either link in firmware, or use both:
- UART5 (PC12 TX, PD2 RX), 3.3 V logic. R54 holds RX high when ramBar is unplugged.
- CAN FD on FDCAN2 (PB13 TX, PB12 RX) through U16, a 3.3 V TCAN334G. R53 terminates this end; ramBar must terminate its end.
  Y1 (8 MHz) is fitted for the CAN clock.
- 5 V to ramBar through F4 (0.5 A hold). U17 is ESD on TX and RX; D12 (NUP2105L) clamps CANH and CANL.



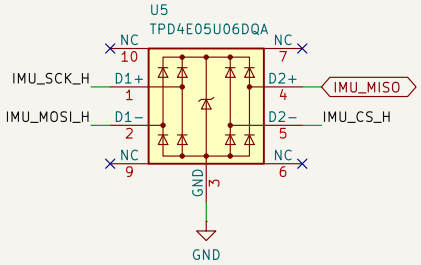
UWAT VEXU		
Sheet: /USB_Jetson/		
File: usb_jetson.kicad_sch		
Title: Centurion Sensor Board		
Size: A4	Date:	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 7/8



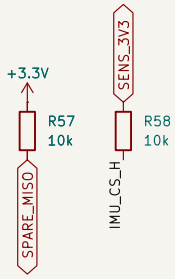




J5 IMU breakout (BMI270 or any SPI IMU): 1 3V3, 2 GND, 3 SCK, 4 MOSI (SDI), 5 MISO (SDO), 6 CS, 7 INT1, 8 INT2. SPI1 on the MCU. Keep the cable under about 30 cm.
3V3 comes from SENS_3V3 (F2), so a shorted cable cannot pull down the main 3.3 V rail.



J15 spare SPI (SPI3): 1 3V3, 2 GND, 3 SCK, 4 MOSI, 5 MISO, 6 CS, 7 INT.
For a plug-in board later, for example a MAX3421E USB host breakout.
A USB host also needs its own switched, current limited 5 V for VBUS.



UWAT VEXU		
Sheet: /IMU/		
File: imu.kicad_sch		
Title: Centurion Sensor Board		
Size: A4	Date:	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 8/8